

Impact of User Mobility on Attentional Tunneling in Handheld AR

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ABSTRACT

Attentional tunneling in augmented reality (AR) refers to a phenomenon where users pay disproportionately high attention to virtual content while ignoring events in the real world. Although many handheld AR applications are deployed in public places where users are expected to be attentive to their surroundings while walking freely, the effect of user mobility on attentional tunneling is not yet ascertained. To investigate the effects of user mobility, we designed a 2x3 study comparing two postures (stationary and walking) with three AR tasks (a non-cognitive baseline task, a working memory task, and a perceptual monitoring task). The results suggest that walking significantly magnifies the tunneling effect, provided the user is engaged with a task that is associated with the virtual content being viewed. We demonstrate the significance of considering user mobility when evaluating AR application usage and provide guidelines for designers to evaluate the application in such scenarios.

CCS CONCEPTS

• **Human-centered computing** → *Mixed / augmented reality*;
Empirical studies in HCI; *Mobile devices*;

KEYWORDS

Attentional Tunneling, Augmented Reality, Mobile Devices, Walking

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1 INTRODUCTION

AR-based mobile applications such as Pokémon GO [14] or Google Maps Live View [1] have been immensely popular, but there are significant concerns regarding attentional issues, such as distracted walking and attentional tunneling, arising from their use [14, 22]. In AR, attentional tunneling refers to a phenomenon where users concentrate disproportionately on the virtual contents while ignoring their actual surroundings [21]. Previous studies have shown that the effect has unintended consequences such as inducing slower

response times [7] and causing individuals to miss events entirely [15, 18].

The outdoor usage of mobile applications on-the-go is known to increase the risk of collisions [4], make people overlook vital safety signs [14], and degrade user performance [9]. Despite the fact that walking is a common activity in several popular AR applications such as Pokémon GO [14] and Google Maps Live View [1], no studies have been conducted to determine whether walking activity has any effect on attentional tunneling. In other words, while we know, for instance, that walking causes users to take longer to perform mobile based web search [9], reading or target selection tasks [19], we do not yet know if walking while using AR applications affects their ability to detect and react to events outside of the task at hand. We believe this is an important research gap to be addressed, since it not only helps us better generalize the findings of previous lab studies to practical scenarios, but also helps establish guidelines for AR app designers to create safe and enjoyable apps for users on the move.

With this goal, we designed a mobile application to evaluate the attentional tunneling effect through reaction times of users to an external event simulation. We opted for a handheld device (HHD) over a head-mounted device because mobile devices with such capabilities are currently the most popular for outdoor AR applications. We performed a 2x3 within subject study with 18 participants, based on two POSTURE variations (sitting versus walking), and three TASK variations (no cognitive load, working memory task, and perceptual monitoring task).

The choice of task is decided based on a related work by Mustonen et al. [12] who contrasted the performance of participants in two separate visual tasks, one based on working memory and the other on perceptual monitoring. We identified that a lot of tasks performed in AR-based applications fall under one of these categories. For example, with the Pokémon GO AR game, functions such as remembering the type and location of Pokémon are related to working memory functions, whereas some others like scanning the real and virtual surroundings in order to catch a wild Pokémon are perceptual monitoring tasks [5, 16, 17]. Lastly, Syiem et al. [21] found that attentional tunneling effect in AR may not be present when the participants are not occupied with tasks associated with the virtual content, i.e., the virtual content is purely for aesthetic purposes (such as virtual landscape or street art). To evaluate if their findings hold true even when the user is on the move, we also included a baseline task consisting of virtual content that did not require any mental processing. Hereafter, we collectively refer to Task 1 and Task 2 as cognitive tasks, in contrast to Baseline task which is a non-cognitive task. Reaction time is used as the primary metric for measuring attentional tunneling, in line with past studies [21, 24].

We were able to reproduce the results obtained by Syiem et al. [21] who found that when seated users were primed with some cognitive task related to the virtual content, they took longer to react to the events. We expanded the scope of their findings to the walking scenario and discovered that the impact of task stayed the same. Between walking and sitting postures, participants took 15 to 20 percent more time to react to events in walking posture, but only for the two cognitive tasks. This indicates that in the presence of cognitive tasks, user mobility significantly amplifies the attentional tunneling effect by increasing the reaction times to external events. Between the two cognitive tasks, we did not find any significant difference, which calls for deeper investigation of the underlying processes and confounding variables involved in AR-based cognitive tasks.

Our research presents empirical evidence for the effects of user mobility on attentional tunneling and calls for deeper investigation of the effect of the tasks, which remains unconfirmed. Since walking significantly amplifies the tunneling effect, we emphasize the need for AR app designers to be especially aware of this attentional issue. Specifically, we recommend that walking while viewing task-relevant virtual material should be done only when it is determined that event detection is not critical to user safety and enjoyment. Information intended for aesthetic appeal and unrelated to a task minimally impacts event recognition, regardless of the posture, so it may be optimized to improve the user's experience. Overall, our study contributes to the growing body of research on attentional tunneling by enhancing our knowledge of this phenomenon and the different circumstances that influence it.

2 RELATED WORK

We briefly contextualize our research within the larger body of literature on attentional tunneling effect in AR.

2.1 Attentional tunneling

Attentional tunneling in AR refers to a phenomenon wherein users place excessive focus on virtual content while neglecting their physical surroundings [24]. It is known to produce various measurable effects, the most common being significant increases in reaction times to events, and a higher proportion of missed events [7, 24]. Syiem et al. [21] found that when performing an AR-based task using HHD, the attentional tunneling effect surfaces when users are required to perform some mental task related to the content being displayed. Most of the studies on attentional tunneling so far have been conducted with participants in a seated or stationary position.

2.2 Outdoor use of mobile apps while walking

Several popular AR apps require users to be moving around with their handheld devices [1, 14]. Clawson et al. [4] explored pedestrian safety aspects of mobile phone usage, and found that user mobility leads to attentional issues such as reduced situational awareness or distracted walking. Pourchon et al. [14] found that users walking while using their smartphone in AR exhibit risky walking behaviours, such as abruptly stop walking on the sidewalk without checking around them, or being less vigilant while crossing the road. Likewise, Yoshiki et al. [26] found that users who engage

in smartphone-based tasks while walking visually confirm their walking path ahead only about 30 percent of the time.

Although various aspects of user behavior and task performance while walking with a mobile phone have been studied in the past, very few studies have attempted to examine how walking affects a user's ability to detect and respond to external events. We attempt to fill this void with our research.

2.3 Effects of cognitive task variations in AR

Nenna et al. [13] engaged the participants with three different AR-based cognitive tasks and found the user performance under each task category varied significantly. While several categories of cognitive processes exist [5], Mustonen et al. [12] found that two particular categories produced distinct effects on the task performance of the users while they were walking and using AR simultaneously, namely working memory based tasks and perceptual monitoring based tasks. Working memory tasks involve holding and manipulating information in the mind over a short period of time [6], such as remembering the skills and moves of Pokémon to use them in a battle. Perceptual monitoring (also called sustained attention) tasks involve actively paying attention to and being aware of sensory information in the environment [10], for example, scanning the environment to locate a random wild Pokémon, or responding to navigation instructions in the live view of AR-based Google Maps. Based on recent review of literature on AR and cognitive loads, Buchner et al. [2] stressed the importance of measuring variations of different types of cognitive loads in future AR-based studies. Keeping this in mind, we introduced two cognitive task variations (working memory and perceptual monitoring) in our study, based on: (i). Our observation that a significant proportion of outdoor AR-based tasks subscribe to one of these categories, (ii). The possibility of correlating our findings with those of previous studies, and (iii). To obtain some preliminary insights into the impact of task variations on the tunneling effect in handheld AR, and expand upon those findings in future studies.

3 METHODOLOGY

We conducted a 2x3 controlled experiment to evaluate the effect of posture (sitting vs walking) and AR task (baseline, working memory, perceptual monitoring) variations on attentional tunneling.

3.1 Participants

18 unpaid participants (male = 11) with corrected or normal vision, aged between 23 and 37 ($M = 26.4$, $SD = 3.41$) took part in the experiment. 7 participants reported to have some prior exposure to AR-based applications on mobile phone.

3.2 Apparatus

We designed and developed an iOS application using Swift programming language and Apple's ARKit, and administered our study through an Apple iPhone 11 (model A2221, screen size 6.1 inch) handheld AR device.

3.3 Experiment Design

We employed a 2 POSTURE x 3 TASK within-subject experimental design, with the following levels:

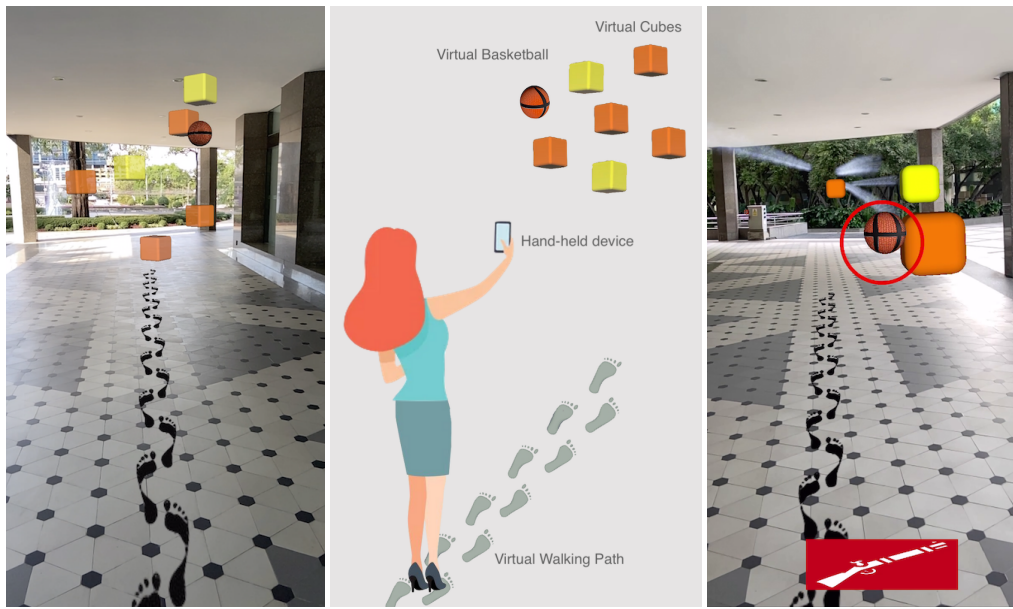


Figure 1: Screen view of Task 1 (Left) and Task 2 (Right) in POSTURE: Walking. (Center) An illustration of our study setup depicting a user holding a HHD. The virtual footprint overlaid on the floor guides the user in the right direction.

- **POSTURE: Sitting** - Participants were asked to sit comfortably on a chair while performing the chosen AR task.
- **POSTURE: Walking** - Participants were asked to walk on a straight, 20 meter long track that was closed to public, and follow a virtual marking on the floor, which was shown in addition to the task related virtual content, as depicted in Figure 1.
- **TASK: Baseline** - The user was presented with virtual content consisting of cubes and spheres overlaid on the video see-through (VST) camera feed. The user performs no mental tasks and makes no interactions with the virtual content. The content presented is analogous to background virtual content in AR applications and games that is meant to improve engagement and aesthetics, such as a graffiti or virtual birds flying in the background, and imposes no cognitive load on users.
- **TASK: Task 1** - A virtual replica of the "Gorilla in our midst" experiment [20, 21] to demonstrate the attentional tunneling effect, wherein participants track a basketball and count its passes amongst a team of moving players. We chose this task not only because it has been used by several attentional tunneling studies previously, but also because it represents a working memory based task [5] where the user must actively remember the number of ball passes.

The user-interface consisted of a full-screen distraction-free VST view with no buttons or obstructions, as shown in Figure 1 (left). Users were presented with six virtual cubes that continuously kept moving to random locations on the screen-space, representing the players passing the ball in the original experiment. Four of these cubes were orange colored, and the remaining two were yellow, representing two teams. A virtual basketball gets passed randomly amongst the six

cubes, and participants were asked to keep a count of ball passes between the orange cubes. The task concludes with a prompt asking the user to enter the count.

- **TASK: Task 2** - The second task was to shoot down incoming virtual objects of a particular color, in AR. This task requires user to stay vigilant and actively scan the display for incoming objects, and is representative of the perceptual monitoring category of tasks [16]. The UI, shown in Figure 1 (right), consisted of a fullscreen VST view with a red circular reticle to assist with aiming, and a button to fire the projectile. The incoming targets all flew in towards the user from the front, but were spread out such that the users had to physically move their small field of view (FoV) HHD to track and shoot the targets.

In summary, for Task 1, the user has to keep track of a virtual ball and maintain a count of the passes in the working memory, whereas for Task 2, the user must monitor the landscape using an AR device to detect incoming virtual objects and respond by shooting them down. AR, by definition, registers virtual content with the physical world's coordinates. To prevent a walking participant from getting too close to the content, and to ensure that the participant had a consistent viewing experience whether seated or on the move, we had to dynamically reposition the virtual content in AR such that the distance between user and the virtual content stays fixed.

Event Trigger: An event trigger represents an unexpected disruption that elicits a specific attentional response. In context of using HHD-based AR apps and games outdoors, such interruptions commonly belong to visual or audio modality, such as approaching vehicles, loud horns, or sirens. Of these two modalities, detecting reaction times for visual events is not straightforward due to the

influence of peripheral vision and other factors that must be compensated for [21]. As a result, we chose to utilize an auditory event trigger, consistent with previous studies [3, 23], by simulating a one-second audio beep played directly from the device.

The dependent variable was the reaction time of users to this event trigger, which was measured in milliseconds. We operationalized the attentional tunneling effect by measuring the time it takes to react to an event, as previous research [7, 21, 24] has shown that the attentional tunneling effect increases reaction times.

3.4 Application Design

The app contained a configuration page for us to select the POSTURE and TASK intended for the trial, and hand over the device to the user. At this point, the app scans the back camera feed to initialize the AR coordinate system, while displaying a start button for the user to initiate the trial. Each trial runs for exactly 30 seconds, during which an auditory event gets triggered at a random duration (between 6-30 seconds), letting user react by tapping anywhere on the screen.

We conducted a pilot run with 4 participants to determine the optimal speeds of the virtual content, the appropriate track length, the right audio levels, and the task duration. We found that it was difficult to keep a participant's attention tunneled to the virtual content over multiple successive trials. A strong learning effect would cause the user to start focusing on the event trigger instead of the virtual content, meaning that neither tunneling of attention nor the element of surprise regarding the unexpected event trigger would be present. Consequently, only one trial was administered to participants in the final study, in line with related previous work [18, 21].

3.5 Procedure

The experiment was conducted in a controlled outdoor setting, consisting of a 20 meter long and 3 meter wide open track that was closed to public. The participants were well rested, asked to read a plain language statement and provide signed informed consent before the experiment began. We selected the right task and posture on the app configuration page, and instructed the participant to tap the screen to begin the trial. Each participant completed the reaction test for all six conditions (within-subject design), with the sequence of trials counterbalanced using a Latin square. We collected 108 reaction time data points in total. The phone was put on flight mode throughout the experiment, and all data was recorded to a CSV file on device.

4 RESULTS

All participants completed each walking posture trial by walking the entire 20 m track within the allotted 30 seconds. The summary statistics of reaction times are as detailed in Table 1. We found that the reaction times for the two cognitive tasks were significantly higher than those of baseline tasks, irrespective of the posture (see Figure 2). Additionally, we also found that for either cognitive task, the reaction time while walking was significantly higher as compared to sitting, as detailed in Table 2. Lastly, we found no significant difference between the two cognitive tasks, as detailed in Table 3.

Table 1: Statistics for Reaction Times, grouped by POSTURE and TASK

POSTURE	TASK	N	MEAN (ms)	SD (ms)
Sitting	Baseline	18	412	16
Sitting	Task 1	18	619	47
Sitting	Task 2	18	618	62
Walking	Baseline	18	443	32
Walking	Task 1	18	731	46
Walking	Task 2	18	736	68

Table 2: Results of post-hoc analysis using pairwise t-test for the effect of POSTURE on each level of TASK.

TASK	POSTURE1	POSTURE2	t	p adj.	Cohen's d
Baseline	Sitting	Walking	1.673	0.730	0.453
Task 1	Sitting	Walking	8.566	<.001***	2.318
Task 2	Sitting	Walking	8.959	<.001***	2.424

Table 3: Results of post-hoc analysis using pairwise t-test for the effect of TASK on each level of POSTURE.

POSTURE	TASK-1	TASK-2	t	p adj.	Cohen's d
Sitting	Task 1	Task 2	0.085	1.0	0.029
Walking	Task 1	Task 2	0.231	1.0	0.078

We tested the data for normality and homogeneity of variances. A Shapiro-Wilk test confirmed normality ($p > 0.05$) for all variables. Mauchly's test of sphericity indicated that the assumption of sphericity has not been violated for TASK ($X^2 = 3.905$, $p = .142$) as well as TASK*POSTURE ($X^2 = 1.671$, $p = .434$). A two-way repeated-measures ANOVA was conducted to compare the effect of POSTURE and TASK on reaction time (in milliseconds) to the events. Significant differences were found in POSTURE ($F_{1,17} = 124.932$, $p < 0.001$, $\eta_p^2 = 0.711$), TASK ($F_{2,34} = 219.869$, $p < 0.001$, $\eta_p^2 = 0.928$), as well as the interaction between the two ($F_{2,34} = 16.657$, $p < 0.001$, $\eta_p^2 = 0.495$).

The post-hoc analysis using pairwise t-test with Bonferroni corrections revealed strong main effect of POSTURE ($p_{bonf} < 0.001$, Cohen's $d = 1.732$). The mean reaction time was 18 percent higher while walking for Task 1, and 19 percent higher for Task 2, as compared to the sitting posture. However, no main effect was found among the two postures for Baseline task ($p_{bonf} < 0.001$, Cohen's $d = 5.024$). We found a strong effect of TASK on reaction times between baseline and Task 1 ($p_{bonf} < 0.001$, Cohen's $d = 4.999$) as well as baseline and Task 2 ($p_{bonf} < 0.001$, Cohen's $d = 5.024$). However, no statistical differences were found between Task 1 and Task 2 ($p_{bonf} = 0.730$ (n.s.)). We also made pairwise comparisons to understand the effects of Task 1 and Task 2 on each level of POSTURE, as detailed in Table 3, and found no significant difference in reaction times among the two cognitive tasks.

5 DISCUSSION

5.1 Effects of user mobility on attentional tunneling

By comparing the reaction times in sitting posture of the two cognitive tasks with baseline task, we were able to reproduce the results from Syiem et al. [21] which state that the virtual content intended for aesthetic appeal but not associated with a task, should have a minimal impact on event detection. In addition, we found the results hold true even in the walking scenario, providing empirical evidence that performing a cognitive task related to the virtual content while walking can result in attentional tunneling. Our findings indicate a substantial increase of 18 to 20 percent in response times for cognitive tasks when comparing the walking posture to the sitting posture, indicating that the walking action magnified the tunneling effect significantly (see Figure 3). Lastly, under the baseline condition, POSTURE variations had no significant effect on reaction times, suggesting that walking action alone does not result in attentional tunneling.

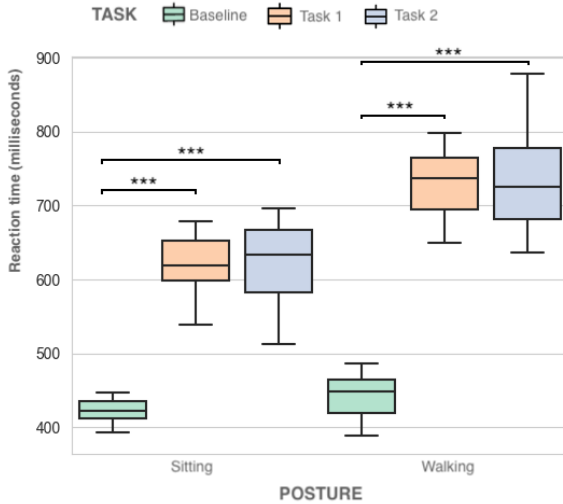


Figure 2: Boxplot of reaction times grouped by POSTURE

To summarize the discussion thus far, we can say that walking does not cause attentional tunneling by itself, but in situations where the user's attention is tunneled through a cognitive task or other means, walking can amplify the effect. These results have consequences on how we assess AR-based outdoor games and navigation apps on mobile devices. In scenarios where the user has to perform some AR-based task while simultaneously remain aware and identify events occurring outside of that task, the designers must consider the influence of user mobility on the users. Our results also show that in the absence of an engaging task, the user's mobility alone is unlikely to impede the user's ability to respond rapidly to an external event. Therefore, AR application designers must examine the following considerations with respect to user mobility and planned application use:

First, regarding to the application's intended use, is the user walking while performing a task associated with the augmented

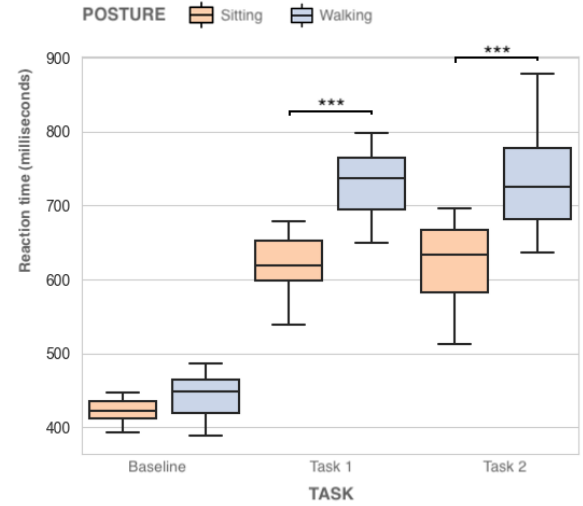


Figure 3: Boxplot of reaction times grouped by TASK

visual content? If yes, the content will capture users' attention, perhaps diminishing the chances of event detection or making users take longer to respond to occurrences, and the walking activity will accentuate this impact even more. However, information intended for aesthetic purposes (such as weather effects like snow and rain, or virtual graffiti) and unrelated to a task should have no impact on event detection, regardless of the posture.

Second, in what context is the user performing the task while walking? App designers may let users view task-relevant augmented content while on the move, only after ensuring that detection of event is no longer critical to user's safety or enjoyment. Designers of AR navigation-based applications or AR based outdoor games must especially consider user attention when reviewing their apps for safety. On the other hand, AR applications deployed in a somewhat secure setting, such as IKEA Place, may afford to be less rigorous in their assessment with regards to user attention.

5.2 Effects of Task 1 versus Task 2

Previous studies [12] have indicated that walking in AR while performing working memory based tasks produces distinct results as compared to perceptual monitoring based tasks, but this did not hold true for our study. The SD of the reaction times for Task 2 was much higher than that of Task 1, as detailed in Table 1, yet pairwise post-hoc comparisons of the two tasks did not reveal any main effects between the two, as detailed in Table 3.

The findings do not necessarily indicate that cognitive task variations have no effect on attentional issues in this context. Since there are hardly any guidelines for designing AR tasks that target specific aspects of cognitive functions, we cannot be certain if the two tasks elicited the desired cognitive effect. Besides, previous studies were conducted on a head-mounted device which has a much wider FoV, whereas we used a handheld device. We call for further research to consider novel variations of the AR-based cognitive tasks, include different types of AR devices, and even try large sample sizes to get a better understanding of this issue.

6 LIMITATIONS AND FUTURE WORK

One important limitation is the lack of standardized cognitive psychology tests such as PEBL [11] designed for AR-specific scenarios. Future work must adopt the standardized tests and tasks for use with AR, while leveraging on the unique features of the technology, for example, designing the AR-equivalent of a Spatial Span test [25] or Spatial Recall test [8] that involves recalling the sequence or location of virtual objects that are placed in the real world.

Another limitation of this study is that it does not account for walking speeds. Static walking techniques like treadmills and walking blocks have been used in the past, but these techniques do not provide meaningful AR experiences. Future studies may consider a combination of in-place walking techniques with a projected AR for background visual motion of a moving environment, to help to overcome the effects of variations in walking speed, while also maintaining ecological validity of the results.

7 CONCLUSION

In keeping with the increasing popularity of AR apps and games in outdoor settings, we investigated the impact of user mobility on the attentional tunneling effect in handheld AR. We found that in presence of cognitive tasks related to virtual content, user mobility amplifies the attentional tunneling effect by increasing the reaction times to external events. We call for additional research to confirm the effects of task variations.

Our findings suggest that information unrelated to a cognitive task minimally impacts event detection, regardless of the posture. Finally, we recommend that walking while viewing task-relevant augmented content may be considered only when the potential impact on detection of important events has been assessed and deemed not crucial to the user's safety and enjoyment.

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